

Ch6 Worked problems

Q1) Need to check; intra pod and inter pod

From the previous page, the minimum local tokens per TPU for FSDP + TP = 113 tokens $\left(\frac{25507}{224}\right)$

Also based on the previous page's question, since the current number of tokens is $\frac{1}{4}$ of the original,

$$\text{Current local tokens per TPU} = \frac{468}{4} = 117 > 113$$

$\therefore$ FSDP + TP barely passes the intra-pod threshold.

Now between the 4 pods, they communicate via the slower

$$\text{DCN connection, the DCN roofline ratio} = \frac{C}{W_{\text{DCN}}} = \frac{4.59 \times 10^{14}}{6.25 \times 10^9}$$

$\approx 73,440 \Rightarrow$ Each pod must have $> 73,440$ tokens. Since each pod has roughly 1M tokens, inter pod connection is completely fine as well.

$\therefore$ Use FSDP + TP mixed, compute-bound

How long it will take to train:

The book already concluded above that 1 pod will take ≈ 44 days,

$\Rightarrow$ 4 pods should take ≈ 11 days.

Q2) Total params = Atten params + MLP params + Vocab params

~~= #~~

$$= L(2DNH + 2DKH) + 3LDF + 2VD$$

(where $L=126$, $D=16384$, $N=128$, $K=8$, $H=128$, $F=53248$, $V=128256$)

$$= 126((2)(16384)(128)^2 + 2(16384)(8)(128))$$

$$+ 3(126)(16384)(53248)$$

$$+ 2(16384)(128256)$$

$$= 71873593344 + 329772957696 + 4202692608$$

$$\approx 405B$$

Training FLOPs per token = 6P

FLOPs per training step = 6PB (P=405B)

Total trng FLOPs for 15T tokens = 6(405B)(15T)

$$= 3.65 \times 10^{25} \text{ FLOPs}$$

2b) Assuming a batch size of 4M tokens, then $B_{pod} = \frac{4194304}{8} = 524288$

• Is inter pod comms. ok? $524288 > 73440 \checkmark$

• Intra? Is tokens per chip $> \frac{2550^2}{2P} = \frac{2550^2}{2(53248)} \approx 61$ tokens per chip.

Given that one pod has 524288 tokens per batch and 8960 chips, each chip receives 58.5 tokens per chip $< 61 \Rightarrow$ we are slightly communication bound within the pod.

(Pure FSDP would require ≈ 850 tokens per chip as calculated in an earlier gn $\Rightarrow$ ~~extremely~~ extremely comms-bound!)

Let's start with the compute time.

$$\text{Compute time @ 40\% MFU} = \frac{3.65 \times 10^{25} \text{ FLOPs}}{8 \times 8960 \times 0.4 \times 459614}$$

≈ 32.1 days.

However, given that we are slightly comms bound for FSDP + TP (58.5 < 61) we need to calculate T_{comms} as well, although I'd believe that 32.1 days would be a reasonable estimate for total time. $\therefore$ I'd estimate 32-33 days